

Question 1 of arXiv:2408.08935 is answered by arXiv:2602.01421

fa_banach_001 literature-status packet

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Status. This is a *literature already answered* packet. It records an exact later answer to one open question from Andrea Garcia, *Greedy algorithms: a review and open problems*, arXiv:2408.08935.

Source question. In Section 3 of arXiv:2408.08935, after defining the Power-Relaxed Greedy Algorithm with relaxation parameter $m^{-\alpha}$, the source asks Question 1: whether it is possible to find $\alpha_0 > 1$ such that

$$\|f - \mathcal{T}_m^r(f)\| \lesssim m^{-\alpha_0/2}, \quad m = 1, 2, \dots,$$

for every $f \in A_1(\mathcal{D})$. This asks whether the exponent in the DeVore–Temlyakov relaxed greedy estimate can be improved by taking $\alpha > 1$ in the power-relaxed variant.

Supporting answer. Pablo M. Berna and Andrea Garcia, *Convergence Analysis of Greedy Algorithms with Adaptive Relaxation in Hilbert Spaces*, arXiv:2602.01421, explicitly restates this source problem as Question 2.3, citing “Greedy algorithms: a review and open problems (2025), Question 1”. Their Theorem 2.4 gives a negative answer: for every $\alpha > 1$, there is a dictionary in $\mathbb{R}^2$ and an element $f \in A_1(\mathcal{D})$ for which the Power-Relaxed Greedy Algorithm fails to have the desired convergence. The proof uses the fact that

$$P_\alpha = \prod_{k=2}^{\infty} \left(1 - \frac{1}{k^\alpha}\right)$$

is strictly positive when $\alpha > 1$, yielding a positive lower bound on the error.

Scope limitations. This identification answers only Question 1 of arXiv:2408.08935. It does not answer Question 2 on alternative step functions, nor Questions 3–5 concerning semi-greedy bases, the isometric 1-almost-greedy versus 1-semi-greedy problem, or semi-greedy renormings.

References.

- A. Garcia, *Greedy algorithms: a review and open problems*, arXiv:2408.08935.
- P. M. Berna and A. Garcia, *Convergence Analysis of Greedy Algorithms with Adaptive Relaxation in Hilbert Spaces*, arXiv:2602.01421.